International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Total Marks: 40 **Mid-sem Exam** Duration: 90 Minutes

Guidelines.

- It is a closed notes/book exam but one A4 cheat sheet is allowed.
- Attempt any 4 questions out of 5.
- If you attempt all 5 five questions, the best 4 will be counted.
- You may freely use the algebraic and order properties of the naturals, integers and rationals, including the properties of absolute value, and the Archimedean property for rationals.

Question (1) [10 Marks]. You may use the following fact without proof: Let $p = (x, y)$ be an ordered pair. Then for all sets X and Y, $p \in X \times Y \iff (x \in X \text{ and } y \in Y)$.

Let A, B, C, D be non-empty sets. Then

(a) **[5 Marks]** Show that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq D$.

(b) **[5 Marks]** Show that $A \times B = C \times D$ if and only if $A = C$ and $B = D$.

Question (2) [10 Marks]. Define two sequences of natural numbers by $p_1 := 1$, $q_1 := 1$ and

$$p_{n+1} := p_n + 2q_n \,, \quad q_{n+1} := p_n + q_n.$$

(a) [5 Marks] Prove that $p_n \geq q_n \geq 1$ for every $n \geq 1$, and that $p_n^2 - 2q_n^2 = (-1)^n$ (an identity between integers).

(b) [5 Marks] Deduce that $q_{n+1} \geq 2q_n$, hence $q_n \geq 2^{n-1}$, hence $q_n \geq n$ for every $n \geq 1$.

Question (3) [10 Marks]. Let $(a_n)_{n\geq 1}$ be a sequence of rationals. For a given rational $\varepsilon > 0$, the sequence $(a_n)_{n\geq 1}$ is said to be eventually ε-steady iff there exists a natural number $N \geq 1$ such that $|a_i - a_j| \leq \varepsilon$ for all $i, j \geq N$.

(a) [5 Marks] For every rational $\varepsilon > 0$, give an example of an eventually ε-steady sequence of rationals which is not a Cauchy sequence. Justify your answer.

(b) [5 Marks] Is the sequence $(a_n)_{n\geq 1}$, where $a_n = (-1)^n$ for all $n \geq 1$, a Cauchy sequence? Justify your answer.

Question (4) [10 Marks]. A sequence $(a_n)_{n\geq 1}$ of rationals is called *anchored* if for every rational $\varepsilon > 0$ there exists natural $N \geq 1$ with

$$|a_j - a_N| \leq \varepsilon \qquad \text{for all } j \geq N.$$

Prove that a sequence of rationals is anchored if and only if it is a Cauchy sequence.

Question (5) [10 Marks]. A function $D : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ is said to be a distance measure on $\mathbb{Q}$ if for all $a, b, c \in \mathbb{Q}$: (1) $D(a,b) \geq 0$ and equality holds iff $a = b$. (2) $D(a,b) = D(b,a)$. (3) $D(a,b) \leq D(a,c) + D(c,b)$. A sequence $(a_n)_{n\geq 1}$ of rationals is called D-Cauchy iff

$$\forall \text{ rational } \varepsilon > 0 \quad \exists \text{ natural } N \geq 1 \text{ such that } \quad D(a_i, a_j) \leq \varepsilon \quad \forall i, j \geq N.$$

Then for $D(a,b) = |a - b|$, we get the usual definition of Cauchy sequences. Now define a function $\phi : \mathbb{Q} \to \mathbb{Q}$ by $\phi(x) = 1/x$ for all nonzero rationals x and $\phi(0) = 0$. Then define $D_\phi(a,b) = |\phi(a) - \phi(b)|$ for all $a, b \in \mathbb{Q}$.

(a) [5 Marks] Prove that D_ϕ is a valid distance measure on $\mathbb{Q}$.

(b) [5 Marks] Give an example of a sequence of rationals that is D_ϕ-Cauchy but not a Cauchy sequence. Justify your answer.